

Course: B.Pharmacy

Subject Name: Pharmaceutical Organic Chemistry- III

Max. Marks: 75

Semester: IV

Subject Code: BP401T

Duration: 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams/figures wherever necessary
3. Figures to right indicate full marks

Q. 1- Objective type question (Answer all the questions)**(10X2) =20**

- i. Define Enantiomers and Diastereoisomerism.
- ii. Draw the structure of Indole. Write its uses.
- iii. What is sequence rule?
- iv. Write down the classification of isomerism.
- v. How will you obtain from 5-nitro thiazole from thiazole?
- vi. What is Schmidt rearrangement?
- vii. Give structure and medicinal uses of Pyrazole.
- viii. Write difference between Stereospecific and Stereoselective reaction.
- ix. Write reaction for Metal hydride reduction (LiAlH_4).
- x. Define Optical activity with example.

Q. 2-Long answer (Answer 2 out of 3)**(10X2) =20**

- i. Give method of preparation, reaction and medicinal uses of Pyrrole and Furan.
- ii. Explain in detail about Clemmensen reduction and Birch reduction reaction.
- iii. What is Confirmation? Describe the details about confirmation of n-butane and cyclohexane.

Q. 3. Short answer (Answer 7 out of 9)**(5X7) =35**

- i. Give synthesis, reaction and medicinal uses of Quinoline and Isoquinoline.
- ii. Write a note on Atropisomerism.
- iii. Comment on relative aromaticity and reactivity of Pyrrole, Furan and Thiophene
- iv. Explain in detail about Dakin reaction.
- v. Define Geometrical isomerism. Write method of determination of geometrical isomerism.
- vi. Write a note on DL system of nomenclature.
- vii. Discuss about Racemic modification and resolution of racemic mixture.
- viii. Outline the basicity of Pyridine.
- ix. Write the methods of preparation of Imidazole.

-----END OF THE PAPER-----